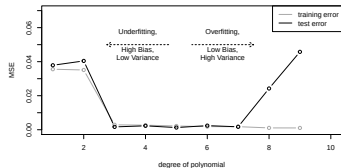
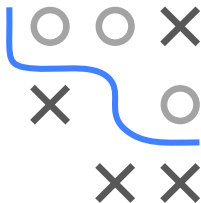


Introduction to Machine Learning

Evaluation Test Error



Learning goals

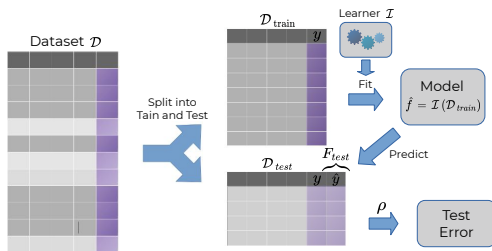
- Understand the definition of test error
- Understand that test error is more reliable than train error
- Bias-Variance analysis of holdout splitting

TEST ERROR AND HOLD-OUT SPLITTING I

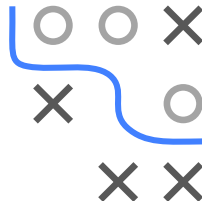
- Simulate prediction on unseen data, to avoid optimistic bias:

$$\rho(\mathbf{y}_{\text{test}}, \mathbf{F}_{\text{test}}) \text{ where } \mathbf{F}_{\text{test}} = \begin{bmatrix} \hat{f}_{\mathcal{D}_{\text{train}}}(\mathbf{x}_{\text{test}}^{(1)}) \\ \vdots \\ \hat{f}_{\mathcal{D}_{\text{train}}}(\mathbf{x}_{\text{test}}^{(m)}) \end{bmatrix}$$

- Partition data, e.g., 2/3 for train and 1/3 for test.

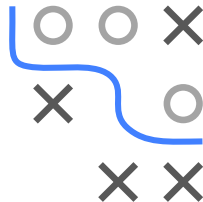
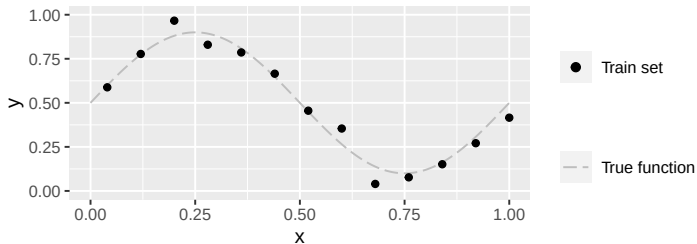


A.k.a. holdout splitting.



EXAMPLE: POLYNOMIAL REGRESSION I

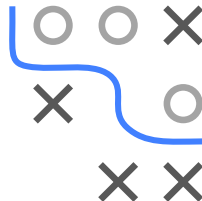
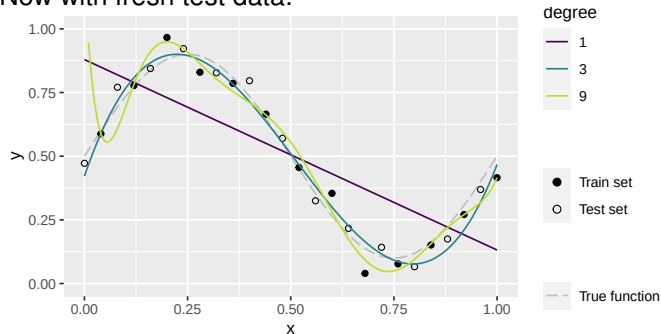
Previous example:



$$f(\mathbf{x} \mid \boldsymbol{\theta}) = \theta_0 + \theta_1 \mathbf{x} + \cdots + \theta_d \mathbf{x}^d = \sum_{j=0}^d \theta_j \mathbf{x}^j.$$

EXAMPLE: POLYNOMIAL REGRESSION II

Now with fresh test data:

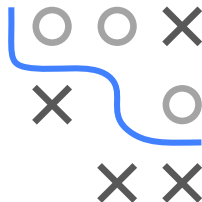
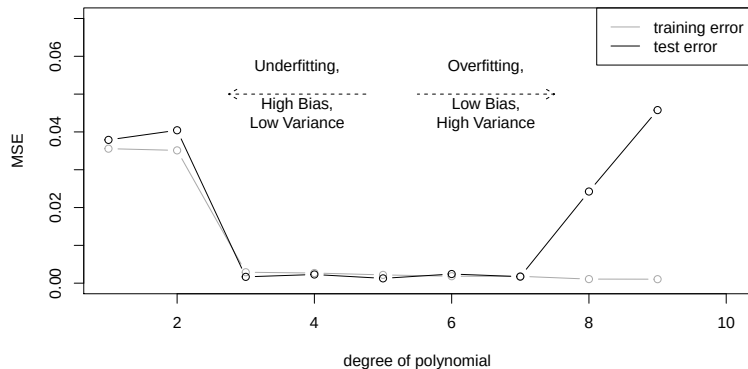


- $d = 1$: MSE = 0.038: clearly underfitting
- $d = 3$: MSE = 0.002: pretty OK
- $d = 9$: MSE = 0.046: clearly overfitting

While train error monotonically decreases in d , test error shows that high- d polynomials overfit.

TEST ERROR I

Let's plot train and test MSE for all d :



Increasing model complexity tends to cause

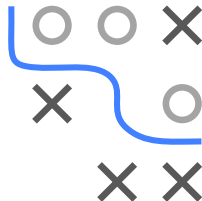
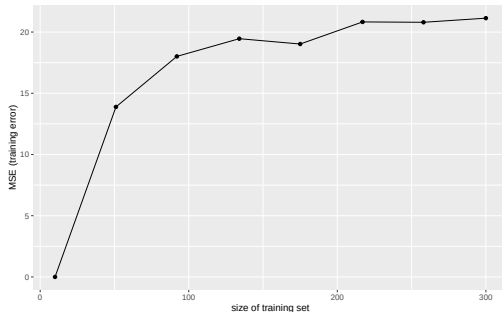
- a decrease in training error, and
- a U-shape in test error
(first underfit, then overfit, sweet-spot in the middle).

TRAINING VS. TEST ERROR I

- Boston Housing data
- Polynomial regression (without interactions)

The training error...

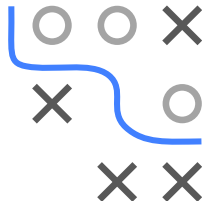
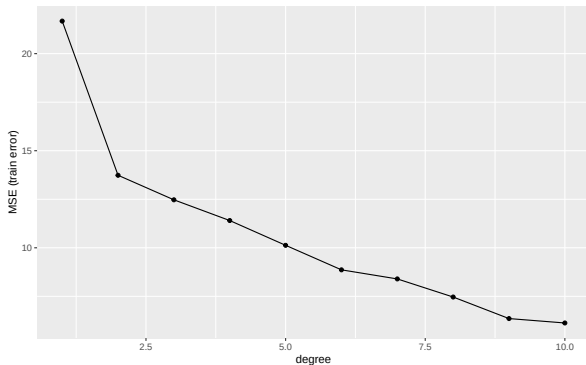
- decreases with smaller training set size as it becomes easier for the model to learn all observed patterns perfectly.



TRAINING VS. TEST ERROR II

The training error...

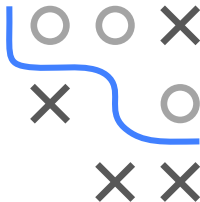
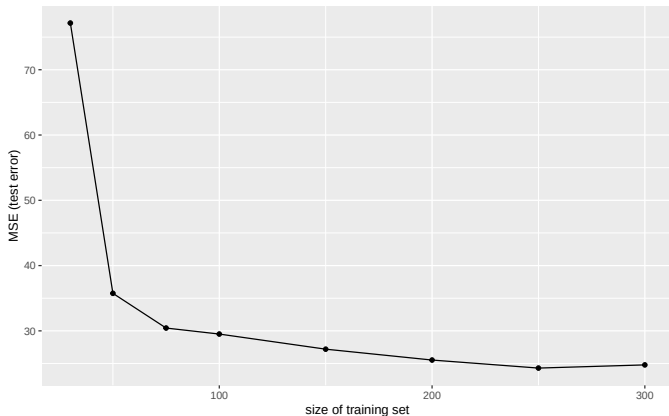
- decreases with increasing model complexity as the model gets better at learning more complex structures.



TRAINING VS. TEST ERROR III

The test error...

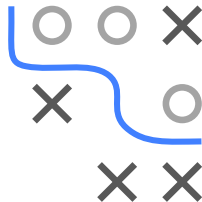
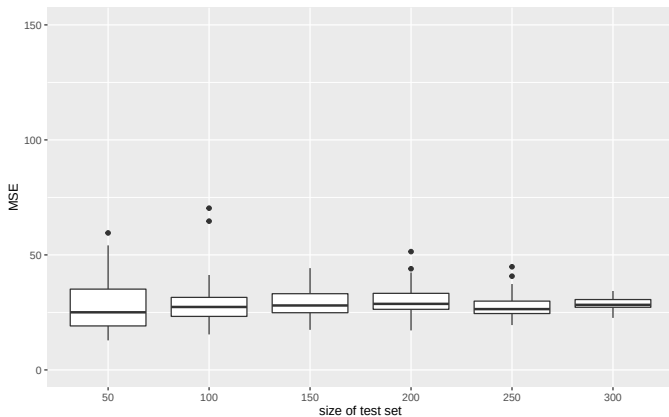
- will typically decrease with larger training set size as the model generalizes better with more data to learn from.



TRAINING VS. TEST ERROR IV

The test error...

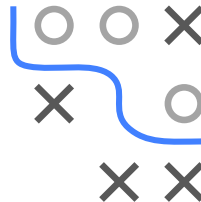
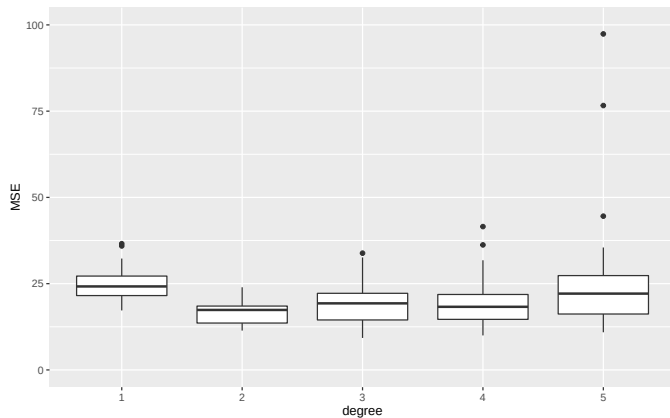
- will have higher variance with smaller test set size.



TRAINING VS. TEST ERROR V

The test error...

- will have higher variance with increasing model complexity.

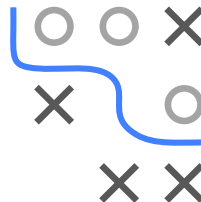
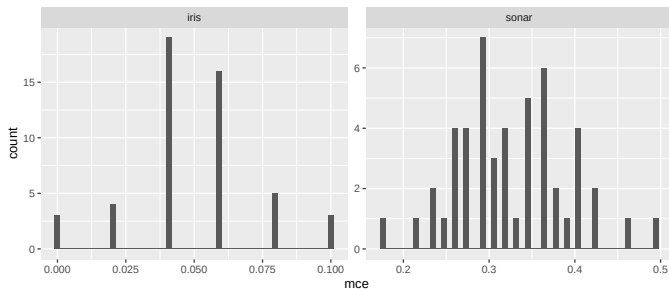


BIAS AND VARIANCE I

- Test error is a good estimator of GE, given a) we have enough data b) test data is representative i.i.d.
- Estimates for smaller test sets can fluctuate considerably – this is why we use resampling in such situations.

Repeated $\frac{2}{3}$ / $\frac{1}{3}$ holdout splits:

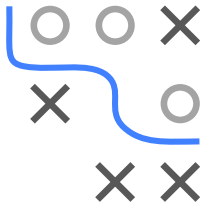
iris ($n = 150$) and sonar ($n = 208$).



BIAS-VARIANCE OF HOLD-OUT – EXPERIMENT I

Hold-out sampling produces a trade-off between **bias** and **variance** that is controlled by split ratio.

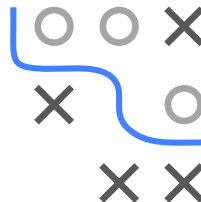
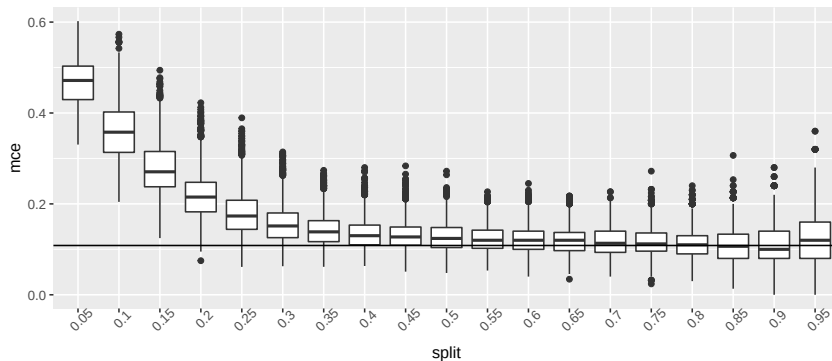
- Smaller training set $\rightarrow$ poor fit, pessimistic bias in $\widehat{GE}$.
- Smaller test set $\rightarrow$ high variance.



Experiment:

- spirals data ($sd = 0.1$), with CART tree.
- Goal: estimate real performance of a model with $|\mathcal{D}_{\text{train}}| = 500$.
- Split rates $s \in \{0.05, 0.10, \dots, 0.95\}$ with $|\mathcal{D}_{\text{train}}| = s \cdot 500$.
- Estimate error on $\mathcal{D}_{\text{test}}$ with $|\mathcal{D}_{\text{test}}| = (1 - s) \cdot 500$.
- 50 repeats for each split rate.
- Get "true" performance by often sampling 500 points, fit learner, then eval on 10^5 fresh points.

BIAS-VARIANCE OF HOLD-OUT – EXPERIMENT II



- Clear pessimistic bias for small training sets – we learn a much worse model than with 500 observations.
- But increase in variance when test sets become smaller.

A 3x3 grid with a blue path starting at the top-left cell (row 1, column 1) and ending at the bottom-right cell (row 3, column 3). The path consists of the following cells: (1,1), (1,2), (2,2), (2,3), and (3,3). The cells (1,3), (2,1), and (3,1) are empty. The cells (1,2), (2,2), and (3,3) contain a gray circle. The cells (2,1), (2,3), and (3,1) contain a gray 'X'.

-
- A line plot showing the Mean Squared Error (MSE) on the y-axis (logarithmic scale, ranging from 0.001 to 0.100) versus the split parameter on the x-axis (ranging from 0.05 to 0.95). The curve starts at approximately 0.12 for split 0.05, decreases to a minimum of about 0.001 at split 0.65, and then increases to approximately 0.004 at split 0.95.
- | split | mse |
|-------|--------|
| 0.05 | 0.12 |
| 0.10 | 0.06 |
| 0.15 | 0.03 |
| 0.20 | 0.015 |
| 0.25 | 0.008 |
| 0.30 | 0.004 |
| 0.35 | 0.0025 |
| 0.40 | 0.0018 |
| 0.45 | 0.0014 |
| 0.50 | 0.0011 |
| 0.55 | 0.0009 |
| 0.60 | 0.0008 |
| 0.65 | 0.0008 |
| 0.70 | 0.0009 |
| 0.75 | 0.0010 |
| 0.80 | 0.0012 |
| 0.85 | 0.0015 |
| 0.90 | 0.0020 |
| 0.95 | 0.0040 |